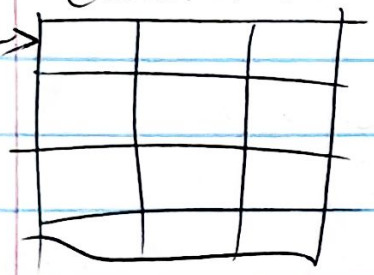
08th JUNE: 26 Monday: SQL: Introduction to SQL Coding!

-> Data: Information we use to make business decisions

Structure — unstructured data

- Stored in a table rxc

Consistent



- When we have data we need to extract insight from data. (Juice from data)

SQL: Structured Query Language;

Programing Language used to extract to get insight from a Structured table - Also in retriving data & information from a structured table.

Student Results

Students ID	name	Marks
101	Nthabi	99
102	Mbali	25
103	Thabo	50
104	Re-Thabo	60

1. Select Statement

- It Also to Select Specific collaum from our table.

2. FROM statement: Allows you specify the name of the table your selecting from;

Select: Name, marks
FROM: Student Results

all these are selecting from what you want

name	marks
Nthobi	99
Mbali	25
Thobo	50
Lethabo	60

the select Statem allows you to specify (you want in your final) the columns Out.

the select Statement Allows You specify the Colaum your final want in the Output.

1. SELECT * select all: Without selecting columns from table without specifying one by one.
2. FR

is A way of selecting

Select *: is A way to select all the Columns from your table without specifying the column one by one
- When you say Select *: you all select the columns in the tab
- Get for all the Column names.

SELECT *
FROM retail_sales_dataset (databricks)

What if You don't want All Columns?

- you separate by a Comma (,) when selecting a specific item column

SELECT date,
Gender,
Age,
Quantity

- selective -

- On the Column you Select specify, You don't put the comma

FROM retail_sales_dataset;

SELECT DISTINCT: different Unique
- to get different values from a specific colum without repeating it.

- We remove duplicates

What can be different?

Syntax: SELECT DISTINCT column name
FROM table_name;

FRUITS orders

Order Product_id	Product name	Price	Revenue
101	Apple	100	
102	Orange	50	
103	Apple	60	
104	Banana	150	

What is Majumbata selling
- Give diffrn value from a specific column
- Apple can't appear any- coz it the same

SELECT DISTINCT prd_name
FROM fruits order

- Go to a specific colum & select a value

Pro_name
Apple
Orang
Banana

How many categories are they bussing trading on? Give me details

SELECT DISTINCT `Product Category` → backtick / Gender
FROM retail_sales_dataset;

Go daddy

CODING: Giving instruction to a computer.

09 June: 26

2 ways of Selecting

1. Select *
From a table; - syntax
→ A way of selecting All the Columns in a from our table

2. Select Distinct: - to Column
From table
No duplicate / specific values

SELECT Column1,
Column2,
Column3

• this way: we want to select specific colounms from our table. WHAT WE WANT.

FROM table_name;

SQL: filtering Statement > Data filtering

-> Able to filter specific data in order to get what we want/data
-> We filter data if: We want to get specific information from our data. (from 2020-7 2030)
- Extraction of data from specific information
- A way to exclude unwanted rows of data that do not need specific criteria.
- we filter based on conditions

How do You Filter data?

-

WHERE Statement.

Syntax: SELECT Column1,
Column2,
Column3
FROM table_name
WHERE Column condition;

Student Rs.

Stu id	name	marks
101	Ntlab	99
102	Ubali	25
103	Thabo	50
104	Lethabo	60

excluded

SELECT *
FROM students results
WHERE marks > 60;

Write an SQL Query to give me Only the names of Student Offband more than 60% at their exam. e.g

Required

Filtered table

Studen id	name	mark
101	Ntlab	99

(in databricks) the moment you specify the number of columns reduced.

Select * —
FROM retail_sales_dataset;
WHERE Gender = 'Male'
(RUN) Results - MALE
- Number of Rows reduces.
Select *
comparsuon operators:

WHERE = 'Male' ~~AND~~ 'Product Catergory' = 'Clothing';

Explain - data filtering in SQL

CORE COMPARISON OPERATORS

These operations are mostly used in ~~Relational~~ Relational Database management System (RDBMS): MySQL, PostgreSQL, SQL Server and Oracle:

COMPARISON OPERATORS

Operator	Descriptive	Example Query
=	Equal to	SELECT * FROM products WHERE price = 50;
<> or !=	Not equal to (<> is the ISO standard)	SELECT * FROM employees WHERE department <> 'HR';
>	Greater than	SELECT * FROM students WHERE marks > 80;
<	Less than	SELECT * FROM inventory WHERE ~~(marks)~~ Stock < 10;
<=	Less than or Equal to	SELECT * FROM flights WHERE delay minutes <= 15;
>=	Greater than or Equal to	SELECT * FROM users WHERE >= 21;

~~4. Arithmetic operators~~

- How to filter data
- SQL operators

Snytax: proflow or follow
- traditions - culture
Gflow of the code
structure

SQL Bitwise operators

Operators	Description
&	Bitwise AND
|	Bitwise OR
^	Bitwise exclusive OR
~	Bitwise NOT

SQL Operators: used also in coding

1. Arithmetic Operators

+	Addition
-	Subtraction
*	Multiplication
/	Division

SELECT *
FROM table-nam
WHERE Colum-nam:

3. Logical Operators

AND	Both conditions must be met
OR	- either or - one condition must be met inclusive statement
Between	must be within the range
IN	must be contained within in the list
NOT	gives the opposite results of what is specified
LIKE	filters) Looks for a pattern

~~Between~~ Lower AND UPPER;

Data filtering filtering statments: filtering statems that filter

-> WHERE statement : based on specific conditions

->We can use SQL operators to build the conditions for filting our data

Syntax: SELECT column 1,
column 2,
column 3;
FROM table_name
WHERE condition;

EXCERSIZE 2

users

user_id	User'name	User city
1	Athabi	JHB
2	David	Cape Town
3	Annie	KZN
4	Kabelo	Pretoria
5	Lerato	Port Elizabeth

how do we filter data?!

QESTION! Write an SQL query to (write) give me the names of users in KZN

SELECT * ~~Column 2~~ Column name Selects all columns that has ppl from durban.
FROM users
WHERE city = 'Durban';

④ Arithmetic operators

- How to filter data
- SQL operators

Syntax: program or follow traditions - culture
Flow of the code structure

SQL Bitwise operators

Operators	Description
&	Bitwise AND
|	Bitwise OR
^	Bitwise exclusive OR
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SQL Operators: used also in coding

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Between	must be within the range
IN	must be contained within in the list
NOT	gives the opposite results of what is specified
LIKE	filters) Looks for a pattern

```
SELECT *
FROM table_name
WHERE Column_name : Between Lower AND UPPER;
```

Data filtering filtering statements: filtering statements that filter
-> WHERE statement : based on specific conditions
-> We can use SQL operators to build the conditions for filtering our data

Syntax : SELECT Column 1,
Column 2,
Column 3;
FROM table_name
WHERE condition;

EXCERSIZE 2

Users

user_id	User'name	User City
1	Athabi	JHB
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How do we filter data?!

QESTION! Write an SQL query to (write) give me the names of users in KZN
SELECT * Column name Selecting all columns that has ppl from durban.
FROM users
WHERE city = 'Durban';

• We used the operator to built the condition =

```
SELECT *
FROM users
WHERE city = 'Durban';
```

(condition; city = 'Durban'; users; column-name; value with = column)

user-id	user-name	city
3	Awele	Durban

Membership_plan

PlanId	plan-name	monthly-price
101	Basic	79
102	Standard	129
103	Premium	199
104	Family	249
105	Mobile	59

6. give out the SQL query that cost more 150 Give the plans the Cost more than 150

```
SELECT * (plan_name, monthly-price)
FROM membership plan
WHERE monthly_price >150
```

```
SELECT *
FROM membership plan
WHERE monthly-price>150
```

P.id	Plan_name	monthly price
103	Premium	199
104	Family	249

(ANS - exercise)

Select all records where the city Column has the value 'Berlin' and the 'Postalcode Column has the value 12209

```
SELECT*
From Customers
Where City = 'Berlin'
```

AND Postalcode = 12209

① OR

Select all records where the City Column has the value 'Berlin' OR 'London'

```
SELECT * FROM Customers
WHERE City = 'Berlin'
OR City = "LONDON'
```

OR

a WHERE clause

② SELECT *
FROM customer table
WHERE city = Berlin or Cardy
= 'Norway'

All costomer can either be from Berlin or Norway.

③ Which SQL Query would Select all spanish Customers whose names start with 'G' or 'R'?

SELECT * FROM customers
WHERE country = Spain
AND (Customers name LIKE 'G'% OR Customername LIKE 'R'%)

so WHERE Country is Either Norway or France

SELECT * Customer
WHERE Country IN ('Nor', 'Franc');

④ Which Query returns all Customers from 'Germany', 'france', or 'UK'?
SELECT * FROM customers
WHERE country IN 'G, F, UK'

SELECT * FROM customers
WHERE country NOT IN ('Mexico, Swed
(Going the opposite)

All customers except those from M and G
(IN)

SELECT * FROM Customers
WHERE country IN ('Norway, France')

Q: USE IN operators to select all the records where Country is either 'Norway' OR 'France'

Use IN operator to Select all the records where Country is NOT 'Norway' and NOT F

SELECT *
FROM customers
WHERE Country NOT IN ('Norway; France

Return all Customers from Country 'Norway; France or USA'
SELECT * FROM Customers
WHERE Country IN ('Norway; France', USA');

BETWEEN

Which SQL Statement Selects all products with a price between 10 & 20?

SELECT * FROM Products
WHERE Price BETWEEN 10 & 20

11 June

SQL Joins

```
SELECT *
FROM Orders
LEFT JOIN Customers
ON Orders.CustomersID = Customers.CustomerID
```

"Insert the missing parts in the join clause to join the 2 tables Orders and Customers', Using the CustomersID field in both tables as the Relationship between the 2 tables

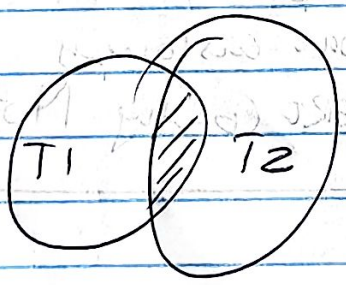
- from the Orders table You join on the left custom where customers ID = customers . customer ID 000.
- key words. [From orders table : customer table] this being the SQL statements : TO join the missing part. You work "ON Orders. Customers = Cus. Cus ID]

②

Join Orders and Customers ON CustomerID

```
SELECT*
FROM Orders
Inner Join Customer ON Orders.CustomerID = Custome=
Custome ID ;
```

Inner join



Inner join

It returns Only rows that have matching values in both tables

```
Select   column name(s)
From     table1
Inner    join  table2
On table1.Column name = table2.Column name.
```

week 3

15 June

SQL Operators

-> to filter out values
-> to exclude rows to our data to use operators
-> Luxury of Filter data

SQL LOGICAL OPERATORS

AND Statement
- can also be used to filter many conditions
- that mean to combine multiples statements / conditions
- Both conditions must be satisfied / met.
- within the give conditions
- we commonly use the AND when having logical conditions for filtering data
- you can specify many conditions as you can.

Syntax

```
SELECT Column1,
       Column2
FROM table_name
WHERE Condition AND Condition2 And Condition3
```

eg

Students

Student id	name	course	Age	City
1	Thabo	Data Analytics	22	Johannesburg
2	Sarah	Data Science	19	Pretoria
3	John	Data Analytics	25	Durban
4	Lerato	Data Analytics	20	Johannesburg
5	Musa	Data Science	23	Johannesburg

Write a SQL query to give me the name of Students living in Johannesburg that are doing Data Analytics?

```
SELECT *
FROM Student
WHERE City AND Courses Analytics
```

```
SELECT *
FROM students
WHERE city = 'Johannesburg'
AND Course = 'Data Analytics';
```

AND =

Outcome

Student ID.	name.	Cource.	Age.	City
1	Thabo	DA	22	JHB
4	Lerato	DA	20	JHB

OR Statement.
- combines multiple conditions
- One of the conditions must be met
- @least meet one of the 2

Students

Stu ID.	name.	Cource	Ag.	City
1	Thabo	DA	22	JhB
2.	Sarah	DS	19	PTA
3	John	~~DA~~	25	DKZN
4	Lerato	DA	23	JHB.
5	Mus	DS	23	JHB

Write a SQL Query to give me the name of Students living in JHB OR the Student Doing DA

```
SELECT *
FROM Students
WHERE City = 'JHB' OR Course = 'DA'
```

Outcome names

1. Thabo
3. Joh
4. Lerato
5. Musa

IN Statement: the way of specifying fitering data by specifying the values of intrest from a specific coloumn.
- point out values from the common coloumn
- there no limit in selecting the values you want

```
SELECT *
FROM table-name
WHERE Column IN('value', 'value2', 'value2');
```

Write an SQL Query to give me ppl living in DBN and Pretoria

```
SELECT name, city
FROM Students
WHERE city IN ('KZN', 'Pretoria');
```

John and Sarah

NOT IN Statement
- Allows us to get opposit Results
- Exclusion statement.

```
SELECT name, city
FROM Students
WHERE city NOT IN ('Durban', Pretoria);
```

Out Come
-> ppl from JHB
1. Thabo
4. Lerato
5. Musa

BETWEEN Statement

-> It allows us to filter data based on a specific range of values

- It is an inclusive statement this means that the lower and upper band of Between Stat are included.

Syntax

```
SELECT Column1,
       Column2,
       Column3;
FROM Students
WHERE Colum BETWEEN lower AND Upper bond
```

Write an SQL Query to give me the name and Age of Student between the Ages of 20 to 23

```
SELECT name, age
FROM Student
WHERE Age= BETWEEN '20 AND 23';
```

Outcome

name	Age
Thabo	22
Lerato	23

SQL (MySQL) programming tool

- Used to get the RDBMS to do things for you
  - Create, Retrieve, Update & delete data
  - Create & manage databases
  - Design & create database tables
  - Perform administration task (security, user management, import/export, etc).

NB:
- Not all RDBMS follow the SQL standard to a T
- The concepts are the same but the implementation may vary.

It a 4 types of Language in one.

1. Data Query L (DQL)
   - Used to query database for information
   - Get information that is already stored there

2. Data Definition Language (DDL)
   - Used to define database schemas (queries, tables, rows)

3. Data Contro Language (D C L)
   - used for controlling access to the database
   - User & permissions management

4. Data Manipulation Language (DML)
   - Used for inserting, updating, deleting data from database

Queries: Set of instruction given to the RDBMS to tell the RDBMS what information you want it to retrieve to you.

- Tons of Data In a DB
- Often Hidden in a Complex schemas
- Goal is to only get the data you need.

```
1 SELECT name, name, employee, age
2 FROM employee
3 WHERE Employee.Salary > 3000;
```

16 June

Sorting Statement:

- way to sort our data by ST : "Order By" statement
- main by 2 way: Assend or dessending order
  (ASC) (DESC)
- Re-arranging
- Sorting doesn't remove/add another new into our data; it just reorange the rows in our data
- Assend - from Lowest = Smallest — Highest = Biggest 9-1 A-Z
- Dessending from higest to lowestest.
  9 - 1
  Z - A

Syntax: SELECT *

```
FROM 'Student' (tablename)
WHERE Condition
ORDER BY Column name ASC;
```

Syntax: DESCENDING

```
SELECT *
FROM table name
WHERE Condition age (a-z)
ORDER BY Column_name DESC;
```

E.g.:

```
SELECT name
       age
FROM Student
ORDER BY age ASC;
```

1.	Thabo	DA	~~20~~	JHB
2.	SARH	DS	19	PTA
3.	John	DA	25	KZN
4.	Lerato	DA	20	JHB
5.	Musa	DS	23	JHB

Out Come:

name	Age
Sara	19
Lerato	20
Thabo	
Musa	22
Musa	23
John	25

Descending:

```
SELECT name
       age
From   student
ORDER BY  age  DESC;
```

Outcome:

name	age
John	25
Nuka	23
Thabo	22
Lerato	20
Sarah	19

- Aggregate functions -

- They unable us to do calculations where they bring a single value.
- An aggregate function is used to perform a calculation on rows of a specific column and returns a single value / single value per / group. • Consolidates the data - Summarize data.

function	Description
1. Count()	Counts the number of rows .Count ids
2. SUM()	Calculate the sum / total of a numeric column
3. AVG()	calculates the Avarage of a numeric column
4. MAX()	Finds the maximum value
5. Min	Finds the minimum value

e.g Employees

emp_id	name	Salary
101	John	20
102	Mary	40
108	David	30

(5) Min

```
SELECT Min(salary)
SELECT Min (Salary) As min-Sal
FROM employees;
```

Outcome: 20

Min-Salary
20

④ SELECT Max(salary) As Max_salary → gives the new column table or Outcome
FROM employee;

Aliases / Nickname
- A descriptive name
- Rename a colum

Outcome

Max_Salary
40

17 June

Grade 10 H

→ Counting the Numbers of learn in a class = 50 (COUNT)
Counts smthing that is simply identified or Unique
each Student had an Id



Student Table

Stu#	name	Subject
101	Mthebi	Science
102	Jeffi	Commerce
103	Mark	commerce
104	Elon	Science
105	Steve	History
106	Donald	Commerce

- we don't count Revenue, we sum the revenue
- We do not sum an Id, rather we count ID

SELECT COUNT(DISTINCT Student id) AS number of Stu
FROM Students;

SELECT
COUNT() (Distinct) id
of Students
From 'Students'

we count ID#
we sum money

Number of Student
6

We are Counting # of Student by Student id.

of Students doing Commerce = 3
of Student doing History = 1
of Student doing Science = 2

GROUP BY statement

- Is a statement that allows us to group our data into categories. / classifications. / groups

Students

Stud	name	Subject
101	Raffi	SC
102	Jeff	com
103	Mark	Com
104	Elon	Scien
105	Steve	Hist
106	Donald	

```
SELECT subject,
SELECT COUNT (DISTINCT student_id) AS number of stud
FROM Student
GROUP BY subject;
```

- Not Aggregated and is commonly used as an tool to find the answer
- is ment to classify
- giving different individual values
- diffent pple in a class
- aggregates
- a clomn that is not aggregated
- we commonly grop by a categorical column (taking it out)

(Sbj (Distinct id) = # Student
GROUP BY Subject
=

Outcome:

Subject	number of Students
Science	2
Commerce	3
Histry	1

Sales

SaleId	Customer_name	Category	Revenue
1	John	Electonics	120
2	Mary	Food	80
3	Elon	Electorics	150
4	Peter	Clothing	100

Write an SQL query to calculate the total revenue for each category

```
SELECT category,
       SUM(revenue) AS Total_revenue
FROM 'Sales'
GROUP BY Category
```

- Here! You do all the calculations as a NEW 'Table'/outcome

Outcome;

Category	total revenue
Electronics	270
Food	80
Clothing	100

You select "Category" You Sum up (revenue) Give it out as a total Rev (being the new column) from sales! then You group it by Category (as it not aggregated) (You GROUP BY what You select being Not the aggr tool)

```
SELECT category,
       Sum(revenue) As Total revenue
FROM 'Sales'
GROUP BY Category
```

- Here! You do all the calculation for a NEW 'Table'/outcome

Outcome;

Category	total revenue
Electronics	270
Food	80
Clothing	100

You select "Category" You Sum up (revenue) Give it out as a total Rev (bring the new column) from sales! then you group it by Category (as it not aggregated) (You GROUP BY what you select being Not the aggregated tool)

18th June 26.

Aggregate functions

- Summurize data
- Performs calculation on multiple rows of a Specific column and returns a simple value/ single per group
- Extract a specific information
- Consilidate data

1. Count() - we count the number of unique value (IDS)
2. SUM () - we Sum the total of the given colum
3. AVG () - we average numerical data (revenue)
4. MIN() - looks for min value in a Colum
5. MAX() - Looks for max value in a Colocum

GROUP BY Statement

```
SELECT category - Column
    Aggregate_function (column) As descriptive column nam
```

FROM table_name
GROUP BY category_column
(we group by the column that is not aggregated) e.g Gender
race
Category

HAVING Statement:
- used to filter data that is aggregated.
We apply the HAVING STATEMENT to data that is aggregated,
- filter group results after using GROUP BY.
(WHERE Filter row befor grouping)
(HAVING Filters groups after group.)

Sales

Stud_id	Customer_name	product	Amount
101	Rofhi	Laptop	5000
101	Rofhi	Mouse	300
102	Abel	keyboard	700
102	Abel	Monitor	2500
103	Ciara	Mouse	300

Write an SQL query to find customers who spend more than 3000?!

```
SELECT customer name, (is aggregated)
       Sum(amount) As total_spent

FROM Sales
GROUP BY customer name;
```

Outcome:

Customer name	total_spent
Rofhi	5300
Abel	3200
Ciara	300

She's still there bcz there NO filtering applied

```
SELECT  Customer_name
        Sum( amount) As total_spent
FROM  Sales
GROUP BY  Customers_name
HAVING  Sum (count)>3000
```

22nd June

Limit

A way of restricting ~~a~~ the number of row you want in your final output.

- Doesn't do any changes but reduce # of rows

Employees

Id	Name
101	Rofhi
102	Mbalo
103	Zum

```
SELECT *
FROM employee
Limit 2
```

Id	name
101	Rofhi
102	Mbalo